RRB JE — Electronics & Allied Engineering Book — Blueprint & Weightage

Source documents (in `RRB/Rrb/`):

- `official-syllabus.pdf` — CEN 05/2025 Annexure-VII C (Electronics & Allied Engineering Exam Group, CBT-2)
- `Electronics-and-Allied-Engineering-Shift-02.pdf` — actual CBT-2 paper (22/04/2025), 100 Q analysed
- Screenshot (Adda247) — CBT-2 marks distribution
- CBT-1 answer keys (19-02-2026 S1/S2) — for CBT-1 pattern reference

CBT-2 Exam Pattern (150 Q / 150 marks, 2 hrs, −1/3 negative)

Section	Questions	Marks
General Awareness	15	15
Physics & Chemistry	15	15
Basics of Computer Applications	10	10
Basics of Environment & Pollution Control	10	10
Technical Abilities (Electronics)	**100**	**100**

> Note: In the analysed Shift-2 paper, Q1–50 covered General Abilities and Q51–100 Technical (shift-level variation exists); the official pattern above is the planning basis.

Technical Abilities — observed distribution (50 technical Q of Shift-2, scaled ×2 to 100)

#	Syllabus Unit (Annexure-VII C)	Observed Q (of 50)	Weight %	Book pages target
2	Electronic Devices & Circuits	8	16%	~40
3	Digital Electronics	8	16%	~40
7	Communication Engineering	8	16%	~40
1	Electronic Components & Materials	5	10%	~25
9	Computer Programming (C/C++, DS, DBMS)	5	10%	~25
5	Microprocessor & Microcontroller (8085/8051)	4	8%	~20
8	Data Communication & Networks	4	8%	~20
10	Basic Electrical Engg. (incl. relays/protection)	4	8%	~20
4	Linear Integrated Circuits (Op-amp, 555, PLL)	2	4%	~12
6	Electronic Measurements	2	4%	~12

Chapter depth follows this weightage — heavier chapters get more solved examples and practice MCQs — but **every syllabus line item is covered**, no topic skipped.

Chapter files

File	Chapter
`chapters/Ch01_Electronic_Components_Materials.md`	Components & Materials
`chapters/Ch02_Electronic_Devices_Circuits.md`	Devices & Circuits
`chapters/Ch03_Digital_Electronics.md`	Digital Electronics
`chapters/Ch04_Linear_Integrated_Circuits.md`	Linear ICs
`chapters/Ch05_Microprocessor_Microcontroller.md`	µP & µC
`chapters/Ch06_Electronic_Measurements.md`	Measurements
`chapters/Ch07_Communication_Engineering.md`	Communication
`chapters/Ch08_Data_Communication_Networks.md`	Data Comm & Networks
`chapters/Ch09_Computer_Programming.md`	Programming
`chapters/Ch10_Basic_Electrical_Engineering.md`	Basic Electrical
`general-abilities/GA01_Physics_NCERT.md`	Physics (NCERT-based)
`general-abilities/GA02_Chemistry_NCERT.md`	Chemistry (NCERT-based)
`general-abilities/GA03_Computer_Applications.md`	Computer basics
`general-abilities/GA04_Environment_Pollution.md`	Environment & Pollution
`general-abilities/GA05_General_Awareness.md`	GA capsule
`pyq-research/CBT1_PYQ_Maths_Reasoning.md`	CBT-1 Maths + Reasoning PYQs

Chapter format standard (every chapter)

1. **Weightage box** — Q expected, PYQ trend
2. **Concept sections** — NCERT-grounded first-principles explanations, every syllabus line item
3. **Worked examples** — numerical + conceptual
4. **PYQ-style solved questions** — including the actual 2025 CBT-2 questions from this unit with solutions
5. **Practice MCQs with answer key**
6. **Rapid-revision formula/fact sheet**